Suggested problems in §14.4: 1-45 odd, 49-53 odd, 57, 59, 69-73 odd

Problem 1: Compute the gradient of the function

$$f(x, y) = y \ln(x) + xy^2$$

at the point $(1, 2)$.

Problem 2: Compute the gradient of the function

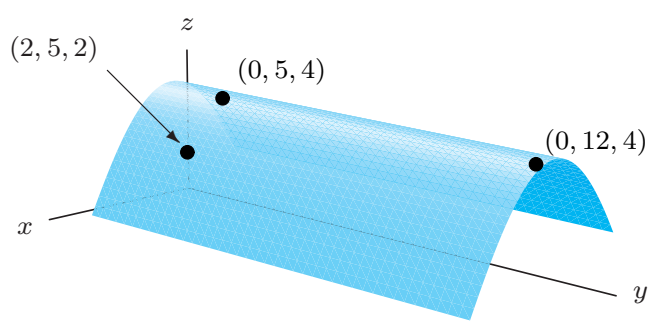
$$f(x, y) = \ln(x^2 + xy)$$

at the point $(4, 1)$.

Problem 3: The surface $z = f(x, y)$ is shown below. Suppose that

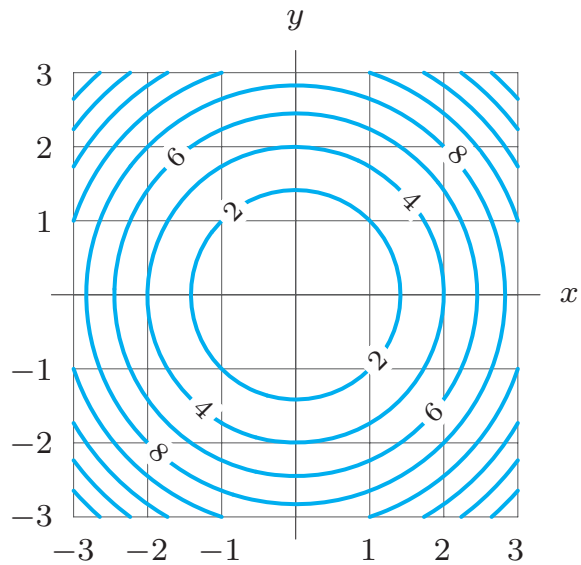
$$\vec{u} = \frac{1}{\sqrt{2}}(\vec{i} + \vec{j}) \quad \text{and} \quad \vec{v} = \frac{1}{\sqrt{2}}(\vec{i} - \vec{j})?$$

Are the following directional derivatives positive, negative, or zero?



(a) $f_{\vec{u}}(2, 5)$
(b) $f_{-\vec{u}}(2, 5)$
(c) $f_{\vec{v}}(2, 5)$
(d) $f_{-\vec{v}}(2, 5)$
(e) $f_{\vec{i}}(2, 5)$
(f) $f_{\vec{j}}(2, 5)$
(g) $f_{\vec{u}}(0, 5)$
(h) $f_{-\vec{u}}(0, 5)$
(i) $f_{\vec{v}}(0, 5)$
(j) $f_{-\vec{v}}(0, 5)$
(k) $f_{\vec{i}}(0, 5)$
(l) $f_{\vec{j}}(0, 5)$

Problem 4: Use the contour diagram of f shown below to determine whether the directional derivative at the indicated point in the indicated direction is positive, negative, or zero.



(a) At $(-2, 2)$, in the direction $\vec{i}$

(b) At $(0, -2)$, in the direction $\vec{j}$

(c) At $(0, -2)$, in the direction $\vec{i} + 2\vec{j}$

(d) At $(0, -2)$, in the direction $\vec{i} - 2\vec{j}$

(e) At $(-1, 1)$, in the direction $\vec{i} + \vec{j}$

(f) At $(-1, 1)$, in the direction $-\vec{i} + \vec{j}$

Problem 5: If $f(x, y) = xy + y^3$ and $\vec{u} = \frac{1}{5}(3\vec{i} - 4\vec{j})$, find $f_{\vec{u}}(1, 2)$.

Problem 6: If $f(x, y) = 3x^2 - 2y^2$, find the instantaneous rate of change of f at $(-3/4, 0)$ in the direction from $(-3/4, 0)$ to $(0, 1)$.